

Making Sense Of The Image Sensors Of The Future

Despite the wide-scale use of conventional CMOS detectors, a possibility exists for the development of image sensors that offer enhanced capabilities regarding increased temporal resolution, better dynamic range, reduced influence of scattering, and more

Dr Matthew Dyson

Conventional image sensors are well-established. And it is expected that in future these will be upgraded with new sensor technologies that will allow us to acquire image based data having more details. This change is mainly influenced by the growing adoption of machine learning for which complex image analysis needs to be performed by computational algorithms. So, the question arises, what will the new image sensing technologies offer? The answer to this question lies in the following points:

- Existing technologies, especially indium-gallium-arsenide (InGaAs) alloy for short-wave infrared (SWIR), are quite expensive and restricted to niche applications. To integrate them into

an autonomous vehicle system, the costs need to be considerably brought down, which the new image sensors promise.

- The availability of increased computer processing power will require more accurate data for analysis and correlation. This is again possible with future high-end image sensors.

On a related note, more advanced sensors working together will surely improve object classification. By comparing the outputs from different sensor types, one will receive accurate insights. Take for example an advanced driver-assistance systems (ADAS) vehicle, which requires lidar, radar, and IR cameras for emergency braking, blind-spot detection, surround view, parking assistance, and much more. Besides this, the advanced sensors' other applications include agriculture, land surveying, industrial monitoring, medical imaging, etc.

So, let's take a look at these emerging image sensor technologies and the specific benefit each delivers.

SWIR image sensing

SWIR is an infrared technology whose wavelength range is 1,000nm to 2,000nm. It can only be detected by InGaAs sensors. In comparison, silicon CMOS image sensors are capable of detecting wavelengths of 400nm to 1,000nm (consisting of visible + near-infrared).

A key benefit of the SWIR image sensor is that it successfully helps in reducing light scatter.

From the graph in Fig. 1, we can deduce that scattering declines as the wavelength of light increases. This is beneficial for autonomous vehicles, ADAS systems, and drones.

The image in the left side of Fig. 2 depicts a person walking through a road on a foggy

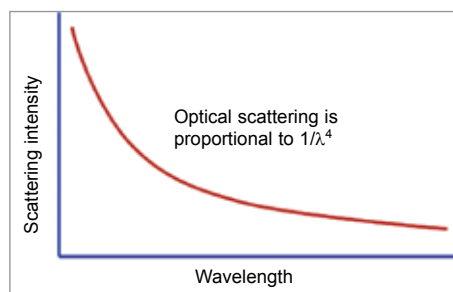


Fig. 1: Declining scattering with increase in wavelength (Credit: IDTechEx)

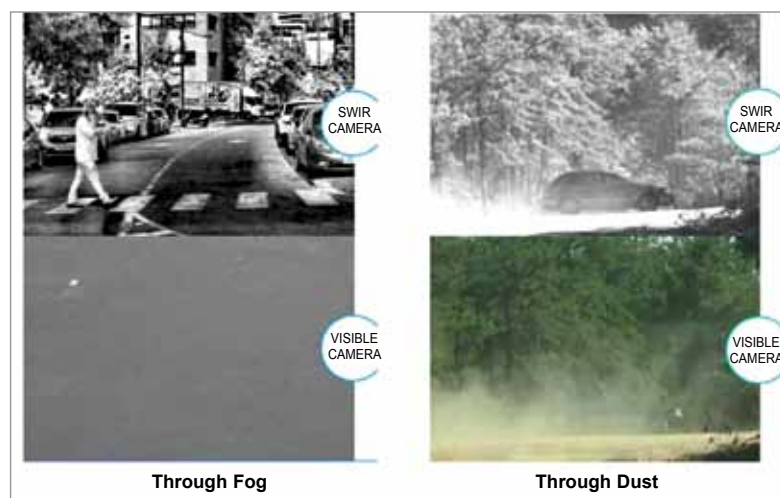


Fig. 2: Comparison of images from SWIR and visible-light cameras (Credit: TriEye)